Cheat Sheet: The pandas DataFrame Object

Preliminaries

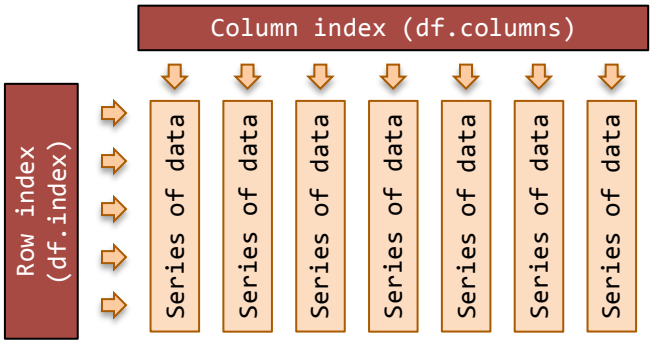
Start by importing these Python modules

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
from pandas import DataFrame, Series
```

Note: these are the recommended import aliases

The conceptual model

DataFrame object: The pandas DataFrame is a two-dimensional table of data with column and row indexes. The columns are made up of pandas Series objects.



Series object: an ordered, one-dimensional array of data with an index. All the data in a Series is of the same data type. Series arithmetic is vectorised after first aligning the Series index for each of the operands.

```
s1 = Series(range(0,4))  # -> 0, 1, 2, 3
s2 = Series(range(1,5))  # -> 1, 2, 3, 4
s3 = s1 + s2             # -> 1, 3, 5, 7
s4 = Series(['a','b'])*3 # -> 'aaa','bbb'
```

The index object: The pandas Index provides the axis labels for the Series and DataFrame objects. It can only contain hashable objects. A pandas Series has one Index; and a DataFrame has two Indexes.

```
# --- get Index from Series and DataFrame
idx = s.index
idx = df.columns    # the column index
idx = df.index      # the row index

# --- some Index attributes
b = idx.is_monotonic_decreasing
b = idx.is_monotonic_increasing
b = idx.has_duplicates
i = idx.nlevels     # multi-level indexes

# --- some Index methods
a = idx.values()    # get as numpy array
l = idx.tolist()    # get as a python list
idx = idx.astype(dtype)# change data type
b = idx.equals(o)   # check for equality
idx = idx.union(o)  # union of two indexes
i = idx.nunique()   # number unique labels
label = idx.min()   # minimum label
label = idx.max()   # maximum label
```

Get your data into a DataFrame

Load a DataFrame from a CSV file

```
df = pd.read_csv('file.csv')# often works
df = pd.read_csv('file.csv', header=0,
      index_col=0, quotechar='"',sep=':',
      na_values = ['na', '-', '.', ''])
```

Note: refer to pandas docs for all arguments

From inline CSV text to a DataFrame

```
from StringIO import StringIO # python2.7
#from io import StringIO      # python 3
data = """, Animal, Cuteness, Desirable
row-1,      dog,    8.7,      True
row-2,      bat,    2.6,      False"""
df = pd.read_csv(StringIO(data),
      header=0, index_col=0,
      skipinitialspace=True)
```

Note: *skipinitialspace=True* allows a pretty layout

Load DataFrames from a Microsoft Excel file

```
# Each Excel sheet in a Python dictionary
workbook = pd.ExcelFile('file.xlsx')
dictionary = {}
for sheet_name in workbook.sheet_names:
    df = workbook.parse(sheet_name)
    dictionary[sheet_name] = df
```

Note: the parse() method takes many arguments like read_csv() above. Refer to the pandas documentation.

Load a DataFrame from a MySQL database

```
import pymysql
from sqlalchemy import create_engine
engine = create_engine('mysql+pymysql://'
    +'USER:PASSWORD@localhost/DATABASE')
df = pd.read_sql_table('table', engine)
```

Data in Series then combine into a DataFrame

```
# Example 1 ...
s1 = Series(range(6))
s2 = s1 * s1
s2.index = s2.index + 2# misalign indexes
df = pd.concat([s1, s2], axis=1)

# Example 2 ...
s3 = Series({'Tom':1, 'Dick':4, 'Har':9})
s4 = Series({'Tom':3, 'Dick':2, 'Mar':5})
df = pd.concat({'A':s3, 'B':s4 }, axis=1)
```

Note: 1st method has in integer column labels
Note: 2nd method does not guarantee col order
Note: index alignment on DataFrame creation

Get a DataFrame from data in a Python dictionary

```
# default --- assume data is in columns
df = DataFrame({
        'col0' : [1.0, 2.0, 3.0, 4.0],
        'col1' : [100, 200, 300, 400]
    })
```

Get a DataFrame from data in a Python dictionary

```
# --- use helper method for data in rows
df = DataFrame.from_dict({ # data by row
        'row0' : {'col0':0, 'col1':'A'},
        'row1' : {'col0':1, 'col1':'B'}
    }, orient='index')

df = DataFrame.from_dict({ # data by row
        'row0' : [1, 1+1j, 'A'],
        'row1' : [2, 2+2j, 'B']
    }, orient='index')
```

Create play/fake data (useful for testing)

```
# --- simple
df = DataFrame(np.random.rand(50,5))

# --- with a time-stamp row index:
df = DataFrame(np.random.rand(500,5))
df.index = pd.date_range('1/1/2006',
       periods=len(df), freq='M')

# --- with alphabetic row and col indexes
import string
import random
r = 52 # note: min r is 1; max r is 52
c = 5
df = DataFrame(np.random.randn(r, c),
       columns = ['col'+str(i) for i in
                    range(c)],
       index = list((string.uppercase +
           string.lowercase)[0:r]))
df['group'] = list(
    ''.join(random.choice('abcd')
        for _ in range(r))
    )
```

Saving a DataFrame

Saving a DataFrame to a CSV file

```
df.to_csv('name.csv', encoding='utf-8')
```

Saving DataFrames to an Excel Workbook

```
from pandas import ExcelWriter
writer = ExcelWriter('filename.xlsx')
df1.to_excel(writer,'Sheet1')
df2.to_excel(writer,'Sheet2')
writer.save()
```

Saving a DataFrame to MySQL

```
import pymysql
from sqlalchemy import create_engine
e = create_engine('mysql+pymysql://' +
    'USER:PASSWORD@localhost/DATABASE')
df.to_sql('TABLE',e, if_exists='replace')
```

Note: if_exists → 'fail', 'replace', 'append'

Saving a DataFrame to a Python dictionary

```
dictionary = df.to_dict()
```

Saving a DataFrame to a Python string

```
string = df.to_string()
```

Note: sometimes may be useful for debugging

Working with the whole DataFrame

Peek at the DataFrame contents

```
df.info()           # index & data types
n = 4
dfh = df.head(n)    # get first n rows
dft = df.tail(n)    # get last n rows
dfs = df.describe() # summary stats cols
top_left_corner_df = df.iloc[:5, :5]
```

DataFrame non-indexing attributes

```
dfT = df.T    # transpose rows and cols
l = df.axes   # list row and col indexes
(r, c) = df.axes  # from above
s = df.dtypes # Series column data types
b = df.empty  # True for empty DataFrame
i = df.ndim   # number of axes (2)
t = df.shape  # (row-count, column-count)
(r, c) = df.shape # from above
i = df.size   # row-count * column-count
a = df.values # get a numpy array for df
```

DataFrame utility methods

```
dfc = df.copy() # copy a DataFrame
dfr = df.rank() # rank each col (default)
dfs = df.sort() # sort each col (default)
dfc = df.astype(dtype) # type conversion
```

DataFrame iteration methods

```
df.iteritems()# (col-index, Series) pairs
df.iterrows() # (row-index, Series) pairs

# example ... iterating over columns
for (name, series) in df.iteritems():
    print('Col name: ' + str(name))
    print('First value: ' +
        str(series.iat[0]) + '\n')
```

Maths on the whole DataFrame (not a complete list)

```
df = df.abs()  # absolute values
df = df.add(o) # add df, Series or value
s = df.count() # non NA/null values
df = df.cummax()  # (cols default axis)
df = df.cummin()  # (cols default axis)
df = df.cumsum()  # (cols default axis)
df = df.cumprod() # (cols default axis)
df = df.diff() # 1st diff (col def axis)
df = df.div(o) # div by df, Series, value
df = df.dot(o) # matrix dot product
s = df.max()   # max of axis (col def)
s = df.mean()  # mean (col default axis)
s = df.median()# median (col default)
s = df.min()   # min of axis (col def)
df = df.mul(o) # mul by df Series val
s = df.sum()   # sum axis (cols default)
```

Note: The methods that return a series default to working on columns.

DataFrame filter/select rows or cols on label info

```
df = df.filter(items=['a', 'b']) # by col
df = df.filter(items=[5], axis=0) #by row
df = df.filter(like='x') # keep x in col
df = df.filter(regex='x') # regex in col
df = df.select(crit=(lambda x:not x%5))#r
```

Note: select takes a Boolean function, for cols: axis=1
Note: filter defaults to cols; select defaults to rows

Working with Columns

A DataFrame column is a pandas Series object

Get column index and labels

```
idx = df.columns           # get col index
label = df.columns[0]      # 1st col label
lst = df.columns.tolist() # get as a list
```

Change column labels

```
df.rename(columns={'old':'new'},
    inplace=True)
df = df.rename(columns={'a':1,'b':'x'})
```

Selecting columns

```
s = df['colName'] # select col to Series
df = df[['colName']] # select col to df
df = df[['a','b']]    # select 2 or more
df = df[['c','a','b']]# change order
s = df[df.columns[0]] # select by number
df = df[df.columns[[0, 3, 4]] # by number
s = df.pop('c') # get col & drop from df
```

Selecting columns with Python attributes

```
s = df.a              # same as s = df['a']
# cannot create new columns by attribute
df.existing_col = df.a / df.b
df['new_col'] = df.a / df.b
```

Trap: column names must be valid identifiers.

Adding new columns to a DataFrame

```
df['new_col'] = range(len(df))
df['new_col'] = np.repeat(np.nan,len(df))
df['random'] = np.random.rand(len(df))
df['index_as_col'] = df.index
df1[['b','c']] = df2[['e','f']]
df3 = df1.append(other=df2)
```

Trap: When adding an indexed pandas object as a new column, only items from the new series that have a corresponding index in the DataFrame will be added. The receiving DataFrame is not extended to accommodate the new series. To merge, see below.
Trap: when adding a python list or numpy array, the column will be added by integer position.

Swap column contents – change column order

```
df[['B', 'A']] = df[['A', 'B']]
```

Dropping columns (mostly by label)

```
df = df.drop('col1', axis=1)
df.drop('col1', axis=1, inplace=True)
df = df.drop(['col1','col2'], axis=1)
s = df.pop('col') # drops from frame
del df['col'] # even classic python works
df.drop(df.columns[0], inplace=True)
```

Vectorised arithmetic on columns

```
df['proportion']=df['count']/df['total']
df['percent'] = df['proportion'] * 100.0
```

Apply numpy mathematical functions to columns

```
df['log_data'] = np.log(df['col1'])
df['rounded'] = np.round(df['col2'], 2)
```

Note: Many more mathematical functions

Columns value set based on criteria

```
df['b']=df['a'].where(df['a']>0,other=0)
df['d']=df['a'].where(df.b!=0,other=df.c)
```

Note: where other can be a Series or a scalar

Data type conversions

```
s =  df['col'].astype(str) # Series dtype
na = df['col'].values      # numpy array
pl = df['col'].tolist()    # python list
```

Note: useful dtypes for Series conversion: int, float, str
Trap: index lost in conversion from Series to array or list

Common column-wide methods/attributes

```
value = df['col'].dtype  # type of data
value = df['col'].size   # col dimensions
value = df['col'].count()# non-NA count
value = df['col'].sum()
value = df['col'].prod()
value = df['col'].min()
value = df['col'].max()
value = df['col'].mean()
value = df['col'].median()
value = df['col'].cov(df['col2'])
s =     df['col'].describe()
s =     df['col'].value_counts()
```

Find index label for min/max values in column

```
label =  df['col1'].idxmin()
label =  df['col1'].idxmax()
```

Common column element-wise methods

```
s = df['col'].isnull()
s = df['col'].notnull() # not isnull()
s = df['col'].astype(float)
s = df['col'].round(decimals=0)
s = df['col'].diff(periods=1)
s = df['col'].shift(periods=1)
s = df['col'].to_datetime()
s = df['col'].fillna(0) # replace NaN w 0
s = df['col'].cumsum()
s = df['col'].cumprod()
s = df['col'].pct_change(periods=4)
s = df['col'].rolling_sum(periods=4,
                          window=4)
```

Note: also rolling_min(), rolling_max(), and many more.

Append a column of row sums to a DataFrame

```
df['Total'] = df.sum(axis=1)
```

Note: also means, mins, maxs, etc.

Multiply every column in DataFrame by Series

```
df = df.mul(s, axis=0) # on matched rows
```

Note: also add, sub, div, etc.

Selecting columns with .loc, .iloc and .ix

```
df = df.loc[:, 'col1':'col2'] # inclusive
df = df.iloc[:, 0:2]          # exclusive
```

Get the integer position of a column index label

```
j = df.columns.get_loc('col_name')
```

Test if column index values are unique/monotonic

```
if df.columns.is_unique: pass # ...
b = df.columns.is_monotonic_increasing
b = df.columns.is_monotonic_decreasing
```

Working with rows

Get the row index and labels

```
idx = df.index              # get row index
label = df.index[0]         # 1st row label
lst = df.index.tolist()     # get as a list
```

Change the (row) index

```
df.index = idx          # new ad hoc index
df.index = range(len(df)) # set with list
df = df.reset_index() # replace old w new
# note: old index stored as a col in df
df = df.reindex(index=range(len(df)))
df = df.set_index(keys=['r1','r2','etc'])
df.rename(index={'old':'new'},
    inplace=True)
```

Adding rows

```
df = original_df.append(more_rows_in_df)
```

Hint: convert to a DataFrame and then append. Both DataFrames should have same column labels.

Dropping rows (by name)

```
df = df.drop('row_label')
df = df.drop(['row1','row2']) # multi-row
```

Boolean row selection by values in a column

```
df = df[df['col2'] >= 0.0]
df = df[(df['col3']>=1.0) |
                        (df['col1']<0.0)]
df = df[df['col'].isin([1,2,5,7,11])]
df = df[~df['col'].isin([1,2,5,7,11])]
df = df[df['col'].str.contains('hello')]
```

Trap: bitwise "or", "and" “not” (ie. | & ~) co-opted to be Boolean operators on a Series of Boolean
Trap: need parentheses around comparisons.

Selecting rows using isin over multiple columns

```
# fake up some data
data = {1:[1,2,3], 2:[1,4,9], 3:[1,8,27]}
df = pd.DataFrame(data)

# multi-column isin
lf = {1:[1, 3], 3:[8, 27]} # look for
f = df[df[list(lf)].isin(lf).all(axis=1)]
```

Selecting rows using an index

```
idx = df[df['col'] >= 2].index
print(df.ix[idx])
```

Select a slice of rows by integer position

[inclusive-from : exclusive-to [: step]]
default start is 0; default end is len(df)

```
df = df[:]        # copy DataFrame
df = df[0:2]      # rows 0 and 1
df = df[-1:]      # the last row
df = df[2:3]      # row 2 (the third row)
df = df[:-1]      # all but the last row
df = df[::2]      # every 2nd row (0 2 ..)
```

Trap: a single integer without a colon is a column label for integer numbered columns.

Select a slice of rows by label/index

[inclusive-from : inclusive–to [: step]]

```
df = df['a':'c'] # rows 'a' through 'c'
```

Trap: doesn't work on integer labelled rows

Append a row of column totals to a DataFrame

```
# Option 1: use dictionary comprehension
sums = {col: df[col].sum() for col in df}
sums_df = DataFrame(sums,index=['Total'])
df = df.append(sums_df)

# Option 2: All done with pandas
df = df.append(DataFrame(df.sum(),
                    columns=['Total']).T)
```

Iterating over DataFrame rows

```
for (index, row) in df.iterrows(): # pass
```

Trap: row data type may be coerced.

Sorting DataFrame rows values

```
df = df.sort(df.columns[0],
                          ascending=False)
df.sort(['col1', 'col2'], inplace=True)
```

Random selection of rows

```
import random as r
k = 20 # pick a number
selection = r.sample(range(len(df)), k)
df_sample = df.iloc[selection, :]
```

Note: this sample is not sorted

Sort DataFrame by its row index

```
df.sort_index(inplace=True) # sort by row
df = df.sort_index(ascending=False)
```

Drop duplicates in the row index

```
df['index'] = df.index # 1 create new col
df = df.drop_duplicates(cols='index',
        take_last=True)# 2 use new col
del df['index']        # 3 del the col
df.sort_index(inplace=True)# 4 tidy up
```

Test if two DataFrames have same row index

```
len(a)==len(b) and all(a.index==b.index)
```

Get the integer position of a row or col index label

```
i = df.index.get_loc('row_label')
```

Trap: index.get_loc() returns an integer for a unique match. If not a unique match, may return a slice or mask.

Get integer position of rows that meet condition

```
a = np.where(df['col'] >= 2) #numpy array
```

Test if the row index values are unique/monotonic

```
if df.index.is_unique: pass # ...
b = df.index.is_monotonic_increasing
b = df.index.is_monotonic_decreasing
```

Working with cells

Selecting a cell by row and column labels

```
value = df.at['row', 'col']
value = df.loc['row', 'col']
value = df['col'].at['row']      # tricky
```

Note: .at[] fastest label based scalar lookup

Setting a cell by row and column labels

```
df.at['row, 'col'] = value
df.loc['row, 'col'] = value
df['col'].at['row'] = value      # tricky
```

Selecting and slicing on labels

```
df = df.loc['row1':'row3', 'col1':'col3']
```

Note: the "to" on this slice is inclusive.

Setting a cross-section by labels

```
df.loc['A':'C', 'col1':'col3'] = np.nan
df.loc[1:2,'col1':'col2']=np.zeros((2,2))
df.loc[1:2,'A':'C']=othr.loc[1:2,'A':'C']
```

Remember: inclusive "to" in the slice

Selecting a cell by integer position

```
value = df.iat[9, 3]         # [row, col]
value = df.iloc[0, 0]        # [row, col]
value = df.iloc[len(df)-1,
                      len(df.columns)-1]
```

Selecting a range of cells by int position

```
df = df.iloc[2:4, 2:4] # subset of the df
df = df.iloc[:5, :5]   # top left corner
s = df.iloc[5, :] # returns row as Series
df = df.iloc[5:6, :] # returns row as row
```

Note: exclusive "to" – same as python list slicing.

Setting cell by integer position

```
df.iloc[0, 0] = value        # [row, col]
df.iat[7, 8] = value
```

Setting cell range by integer position

```
df.iloc[0:3, 0:5] = value
df.iloc[1:3, 1:4] = np.ones((2, 3))
df.iloc[1:3, 1:4] = np.zeros((2, 3))
df.iloc[1:3, 1:4] = np.array([[1, 1, 1],
                              [2, 2, 2]])
```

Remember: exclusive-to in the slice

.ix for mixed label and integer position indexing

```
value = df.ix[5, 'col1']
df = df.ix[1:5, 'col1':'col3']
```

Views and copies

From the manual: Setting a copy can cause subtle errors. The rules about when a view on the data is returned are dependent on NumPy. Whenever an array of labels or a Boolean vector are involved in the indexing operation, the result will be a copy.

In summary: indexes and addresses

In the main, these notes focus on the simple, single level Indexes. Pandas also has a hierarchical or multi-level Indexes (aka the MultiIndex).

A DataFrame has two Indexes

- Typically, the column index (df.columns) is a list of strings (observed variable names) or (less commonly) integers (the default is numbered from 0 to length-1)
- Typically, the row index (df.index) might be:
  - Integers - for case or row numbers (default is numbered from 0 to length-1);
  - Strings – for case names; or
  - DatetimeIndex or PeriodIndex – for time series data (more below)

Indexing

```
# --- selecting columns
s =  df['col_label']     # scalar
df = df[['col_label']]   # one item list
df = df[['L1', 'L2']]    # many item list
df = df[index]           # pandas Index
df = df[s]               # pandas Series

# --- selecting rows
df = df['from':'inc_to']# label slice
df = df[3:7]             # integer slice
df = df[df['col'] > 0.5]# Boolean Series

df = df.loc['label']     # single label
df = df.loc[container]   # lab list/Series
df = df.loc['from':'to']# inclusive slice
df = df.loc[bs]          # Boolean Series
df = df.iloc[0]          # single integer
df = df.iloc[container] # int list/Series
df = df.iloc[0:5]        # exclusive slice
df = df.ix[x]            # loc then iloc

# --- select DataFrame cross-section
#     r and c can be scalar, list, slice
df.loc[r, c] # label accessor (row, col)
df.iloc[r, c]# integer accessor
df.ix[r, c]  # label access int fallback
df[c].iloc[r]# chained – also for .loc

# --- select cell
#     r and c must be label or integer
df.at[r, c]  # fast scalar label accessor
df.iat[r, c] # fast scalar int accessor
df[c].iat[r] # chained – also for .at

# --- indexing methods
v = df.get_value(r, c)  # get by row, col
df = df.set_value(r,c,v)# set by row, col
df = df.xs(key, axis) # get cross-section
df = df.filter(items, like, regex, axis)
df = df.select(crit, axis)
```

Note: the indexing attributes (.loc, .iloc, .ix, .at .iat) can be used to get and set values in the DataFrame.
Note: the .loc, iloc and .ix indexing attributes can accept python slice objects. But .at and .iat do not.
Note: .loc can also accept Boolean Series arguments
Avoid: chaining in the form df[col_indexer][row_indexer]
Trap: label slices are inclusive, integer slices exclusive.

Joining/Combining DataFrames

Three ways to join two DataFrames:

- merge (a database/SQL-like join operation)
- concat (stack side by side or one on top of the other)
- combine_first (splice the two together, choosing values from one over the other)

Merge on indexes

```
df_new = pd.merge(left=df1, right=df2,
          how='outer', left_index=True,
          right_index=True)
```

How: 'left', 'right', 'outer', 'inner'
How: outer=union/all; inner=intersection

Merge on columns

```
df_new = pd.merge(left=df1, right=df2,
          how='left', left_on='col1',
          right_on='col2')
```

Trap: When joining on columns, the indexes on the passed DataFrames are ignored.
Trap: many-to-many merges on a column can result in an explosion of associated data.

Join on indexes (another way of merging)

```
df_new = df1.join(other=df2, on='col1',
                  how='outer')
df_new = df1.join(other=df2,on=['a','b'],
                  how='outer')
```

Note: DataFrame.join() joins on indexes by default. DataFrame.merge() joins on common columns by default.

Simple concatenation is often the best

```
df=pd.concat([df1,df2],axis=0)#top/bottom
df = df1.append([df2, df3])   #top/bottom
df=pd.concat([df1,df2],axis=1)#left/right
```

Trap: can end up with duplicate rows or cols
Note: concat has an ignore_index parameter

Combine_first

```
df = df1.combine_first(other=df2)

# multi-combine with python reduce()
df = reduce(lambda x, y:
        x.combine_first(y),
            [df1, df2, df3, df4, df5])
```

Uses the non-null values from df1. The index of the combined DataFrame will be the union of the indexes from df1 and df2.

Groupby: Split-Apply-Combine

The pandas "groupby" mechanism allows us to split the data into groups, apply a function to each group independently and then combine the results.

Grouping

```
gb = df.groupby('cat') # by one columns
gb = df.groupby(['c1','c2']) # by 2 cols
gb = df.groupby(level=0) # multi-index gb
gb = df.groupby(level=['a','b']) # mi gb
print(gb.groups)
```

Note: groupby() returns a pandas groupby object
Note: the groupby object attribute .groups contains a dictionary mapping of the groups.
Trap: NaN values in the group key are automatically dropped – there will never be a NA group.

Iterating groups – usually not needed

```
for name, group in gb:
    print (name)
    print (group)
```

Selecting a group

```
dfa = df.groupby('cat').get_group('a')
dfb = df.groupby('cat').get_group('b')
```

Applying an aggregating function

```
# apply to a column ...
s = df.groupby('cat')['col1'].sum()
s = df.groupby('cat')['col1'].agg(np.sum)
# apply to the every column in DataFrame
s = df.groupby('cat').agg(np.sum)
df_summary = df.groupby('cat').describe()
df_row_1s = df.groupby('cat').head(1)
```

Note: aggregating functions reduce the dimension by one – they include: mean, sum, size, count, std, var, sem, describe, first, last, min, max

Applying multiple aggregating functions

```
gb = df.groupby('cat')

# apply multiple functions to one column
dfx = gb['col2'].agg([np.sum, np.mean])
# apply to multiple fns to multiple cols
dfy = gb.agg({
    'cat': np.count_nonzero,
    'col1': [np.sum, np.mean, np.std],
    'col2': [np.min, np.max]
})
```

Note: gb['col2'] above is shorthand for df.groupby('cat')['col2'], without the need for regrouping.

Transforming functions

```
# transform to group z-scores, which have
# a group mean of 0, and a std dev of 1.
zscore = lambda x: (x-x.mean())/x.std()
dfz = df.groupby('cat').transform(zscore)

# replace missing data with group mean
mean_r = lambda x: x.fillna(x.mean())
dfm = df.groupby('cat').transform(mean_r)
```

Note: can apply multiple transforming functions in a manner similar to multiple aggregating functions above,

Applying filtering functions

Filtering functions allow you to make selections based on whether each group meets specified criteria

```
# select groups with more than 10 members
eleven = lambda x: (len(x['col1']) >= 11)
df11 = df.groupby('cat').filter(eleven)
```

Group by a row index (non-hierarchical index)

```
df = df.set_index(keys='cat')
s = df.groupby(level=0)['col1'].sum()
dfg = df.groupby(level=0).sum()
```

Pivot Tables

Pivot

Pivot tables move from long format to wide format data

```
df = DataFrame(np.random.rand(100,1))
df.columns = ['data'] # rename col
df.index = pd.period_range('3/3/2014',
       periods=len(df), freq='M')
df['year'] = df.index.year
df['month'] = df.index.month

# pivot to wide format
df = df.pivot(index='year',
    columns='month', values='data')

# melt to long format
dfm = df
dfm['year'] = dfm.index
dfm = pd.melt(df, id_vars=['year'],
    var_name='month', value_name='data')

# unstack to long format
# reset index to remove multi-level index
dfu=df.unstack().reset_index(name='data')
```

Value counts

```
s = df['col1'].value_counts()
```

Working with dates, times and their indexes

Dates and time – points and spans

With its focus on time-series data, pandas has a suite of tools for managing dates and time: either as a point in time (a Timestamp) or as a span of time (a Period).

```
t = pd.Timestamp('2013-01-01')
t = pd.Timestamp('2013-01-01 21:15:06')
t = pd.Timestamp('2013-01-01 21:15:06.7')
p = pd.Period('2013-01-01', freq='M')
```

Note: Timestamps should be in range 1678 and 2261 years. (Check Timestamp.max and Timestamp.min).

A Series of Timestamps or Periods

```
ts = ['2015-04-01 13:17:27',
        '2014-04-02 13:17:29']

# Series of Timestamps (good)
s = pd.to_datetime(pd.Series(ts))

# Series of Periods (often not so good)
s = pd.Series( [pd.Period(x, freq='M')
        for x in ts] )
s = pd.Series(
        pd.PeriodIndex(ts,freq='S'))
```

Note: While Periods make a very useful index; they may be less useful in a Series.

From non-standard strings to Timestamps

```
t = ['09:08:55.7654-JAN092002',
        '15:42:02.6589-FEB082016']
s = pd.Series(pd.to_datetime(t,
        format="%H:%M:%S.%f-%b%d%Y"))
```

Also: %B = full month name; %m = numeric month; %y = year without century; and more …

Dates and time – stamps and spans as indexes

An index of Timestamps is a DatetimeIndex.
An index of Periods is a PeriodIndex.

```
date_strs = ['2014-01-01', '2014-04-01',
             '2014-07-01', '2014-10-01']

dti = pd.DatetimeIndex(date_strs)

pid = pd.PeriodIndex(date_strs, freq='D')
pim = pd.PeriodIndex(date_strs, freq='M')
piq = pd.PeriodIndex(date_strs, freq='Q')

print (pid[1] - pid[0])  # 90 days
print (pim[1] - pim[0])  #  3 months
print (piq[1] - piq[0])  #  1 quarter

time_strs = ['2015-01-01 02:10:40.12345',
             '2015-01-01 02:10:50.67890']
pis = pd.PeriodIndex(time_strs, freq='U')

df.index = pd.period_range('2015-01',
        periods=len(df), freq='M')

dti = pd.to_datetime(['04-01-2012'],
  dayfirst=True) # Australian date format
pi = pd.period_range('1960-01-01',
  '2015-12-31', freq='M')
```

Hint: unless you are working in less than seconds, prefer PeriodIndex over DateTimeImdex.

Period frequency constants (not a complete list)

Name	Description
U	Microsecond
L	Millisecond
S	Second
T	Minute
H	Hour
D	Calendar day
B	Business day
W-{MON, TUE, …}	Week ending on …
MS	Calendar start of month
M	Calendar end of month
QS-{JAN, FEB, …}	Quarter start with year starting (QS – December)
Q-{JAN, FEB, …}	Quarter end with year ending (Q – December)
AS-{JAN, FEB, …}	Year start (AS - December)
A-{JAN, FEB, …}	Year end (A - December)

From DatetimeIndex to Python datetime objects

```
dti = pd.DatetimeIndex(pd.date_range(
  start='1/1/2011', periods=4, freq='M'))
s = Series([1,2,3,4], index=dti)
na = dti.to_pydatetime()      #numpy array
na = s.index.to_pydatetime() #numpy array
```

Frome Timestamps to Python dates or times

```
df['date'] = [x.date() for x in df['TS']]
df['time'] = [x.time() for x in df['TS']]
```

Note: converts to datatime.date or datetime.time. But does not convert to datetime.datetime.

From DatetimeIndex to PeriodIndex and back

```
df = DataFrame(np.random.randn(20,3))
df.index = pd.date_range('2015-01-01',
       periods=len(df), freq='M')
dfp = df.to_period(freq='M')
dft = dfp.to_timestamp()
```

Note: from period to timestamp defaults to the point in time at the start of the period.

Working with a PeriodIndex

```
pi = pd.period_range('1960-01','2015-12',
                     freq='M')
na = pi.values  # numpy array of integers
lp = pi.tolist() # python list of Periods
sp = Series(pi)# pandas Series of Periods
ss = Series(pi).astype(str) # S of strs
ls = Series(pi).astype(str).tolist()
```

Get a range of Timestamps

```
dr = pd.date_range('2013-01-01',
    '2013-12-31', freq='D')
```

Error handling with dates

```
# 1st example returns string not Timestamp
t = pd.to_datetime('2014-02-30')
# 2nd example returns NaT (not a time)
t = pd.to_datetime('2014-02-30',
      coerce=True)
# NaT like NaN tests True for isnull()
b = pd.isnull(t) # --> True
```

The tail of a time-series DataFrame

```
df = df.last("5M") # the last five months
```

Upsampling and downsampling

```
# upsample from quarterly to monthly
pi = pd.period_range('1960Q1',
        periods=220, freq='Q')
df = DataFrame(np.random.rand(len(pi),5),
        index=pi)
dfm = df.resample('M', convention='end')
# use ffill or bfill to fill with values

# downsample from monthly to quarterly
dfq = dfm.resample('Q', how='sum')
```

Time zones

```
t = ['2015-06-30 00:00:00',
      '2015-12-31 00:00:00']
dti = pd.to_datetime(t
      ).tz_localize('Australia/Canberra')
dti = dti.tz_convert('UTC')
ts = pd.Timestamp('now',
      tz='Europe/London')

# get a list of all time zones
import pyzt
for tz in pytz.all_timezones:
    print tz
```

Note: by default, Timestamps are created without time zone information.

Row selection with a time-series index

```
# start with the play data above
idx = pd.period_range('2015-01',
        periods=len(df), freq='M')
df.index = idx

february_selector = (df.index.month == 2)
february_data = df[february_selector]

q1_data = df[(df.index.month >= 1) &
   (df.index.month <= 3)]

mayornov_data = df[(df.index.month == 5)
   | (df.index.month == 11)]

totals = df.groupby(df.index.year).sum()
```

Also: year, month, day [of month], hour, minute, second, dayofweek [Mon=0 .. Sun=6], weekofmonth, weekofyear [numbered from 1], week starts on Monday], dayofyear [from 1], …

The Series.dt accessor attribute

DataFrame columns that contain datetime-like objects can be manipulated with the .dt accessor attribute

```
t = ['2012-04-14 04:06:56.307000',
     '2011-05-14 06:14:24.457000',
     '2010-06-14 08:23:07.520000']

# a Series of time stamps
s = pd.Series(pd.to_datetime(t))
print(s.dtype)       # datetime64[ns]
print(s.dt.second)   # 56, 24,  7
print(s.dt.month)    #  4,  5,  6

# a Series of time periods
s = pd.Series(pd.PeriodIndex(t,freq='Q'))
print(s.dtype)       # datetime64[ns]
print(s.dt.quarter) #  2,  2,  2
print(s.dt.year)    #  2012, 2011, 2010
```

Working with missing and non-finite data

Working with missing data

Pandas uses the not-a-number construct (np.nan and float('nan')) to indicate missing data. The Python None can arise in data as well. It is also treated as missing data; as is the pandas not-a-time construct (pandas.NaT).

Missing data in a Series

```
s = Series( [8,None,float('nan'),np.nan])
           #[8,     NaN,   NaN,   NaN]
s.isnull() #[False, True,  True,  True]
s.notnull()#[True,  False, False, False]
s.fillna(0)#[8,     0,     0,     0]
```

Missing data in a DataFrame

```
df = df.dropna() # drop all rows with NaN
df = df.dropna(axis=1) # same for cols
df=df.dropna(how='all') #drop all NaN row
df=df.dropna(thresh=2) # drop 2+ NaN in r
# only drop row if NaN in a specified col
df = df.dropna(df['col'].notnull())
```

Recoding missing data

```
df.fillna(0, inplace=True) # np.nan → 0
s = df['col'].fillna(0)    # np.nan → 0
df = df.replace(r'\s+', np.nan,
      regex=True) # white space → np.nan
```

Non-finite numbers

With floating point numbers, pandas provides for positive and negative infinity.

```
s = Series([float('inf'), float('-inf'),
     np.inf, -np.inf])
```

Pandas treats integer comparisons with plus or minus infinity as expected.

Testing for finite numbers

(using the data from the previous example)

```
b = np.isfinite(s)
```

Working with Categorical Data

Categorical data

The pandas Series has an R factors-like data type for encoding categorical data.

```
s = Series(['a','b','a','c','b','d','a'],
                        dtype='category')
df['B'] = df['A'].astype('category')
```

Note: the key here is to specify the "category" data type.
Note: categories will be ordered on creation if they are sortable. This can be turned off. See ordering below.

Convert back to the original data type

```
s = Series(['a','b','a','c','b','d','a'],
                        dtype='category')
s = s.astype('string')
```

Ordering, reordering and sorting

```
s = Series(list('abc'), dtype='category')
print (s.cat.ordered)
s=s.cat.reorder_categories(['b','c','a'])
s = s.sort()
s.cat.ordered = False
```

Trap: category must be ordered for it to be sorted

Renaming categories

```
s = Series(list('abc'), dtype='category')
s.cat.categories = [1, 2, 3] # in place
s = s.cat.rename_categories([4,5,6])
# using a comprehension ...
s.cat.categories = ['Group ' + str(i)
    for i in s.cat.categories]
```

Trap: categories must be uniquely named

Adding new categories

```
s = s.cat.add_categories([4])
```

Removing categories

```
s = s.cat.remove_categories([4])
s.cat.remove_unused_categories() #inplace
```

Working with strings

Working with strings

```
# assume that df['col'] is series of
strings
s = df['col'].str.lower()
s = df['col'].str.upper()
s = df['col'].str.len()

# the next set work like Python
df['col'] += 'suffix'        # append
df['col'] *= 2               # duplicate
s = df['col1'] + df['col2'] # concatenate
```

Most python string functions are replicated in the pandas DataFrame and Series objects.

Regular expressions

```
s = df['col'].str.contains('regex')
s = df['col'].str.startswith('regex')
s = df['col'].str.endswith('regex')
s = df['col'].str.replace('old', 'new')
df['b'] = df.a.str.extract('(pattern)')
```

Note: pandas has many more regex methods.

Basic Statistics

Summary statistics

```
s = df['col1'].describe()
df1 = df.describe()
```

DataFrame – key stats methods

```
df.corr()      # pairwise correlation cols
df.cov()       # pairwise covariance cols
df.kurt()      # kurtosis over cols (def)
df.mad()       # mean absolute deviation
df.sem()       # standard error of mean
df.var()       # variance over cols (def)
```

Value counts

```
s = df['col1'].value_counts()
```

Cross-tabulation (frequency count)

```
ct = pd.crosstab(index=df['a'],
        cols=df['b'])
```

Quantiles and ranking

```
quants = [0.05, 0.25, 0.5, 0.75, 0.95]
q = df.quantile(quants)
r = df.rank()
```

Histogram binning

```
count, bins = np.histogram(df['col1'])
count, bins = np.histogram(df['col'],
                bins=5)
count, bins = np.histogram(df['col1'],
                bins=[-3,-2,-1,0,1,2,3,4])
```

Regression

```
import statsmodels.formula.api as sm
result = sm.ols(formula="col1 ~ col2 +
          col3", data=df).fit()
print (result.params)
print (result.summary())
```

Smoothing example using rolling_apply

```
k3x5 = np.array([1,2,3,3,3,2,1]) / 15.0
s = pd.rolling_apply(df['col1'],
       window=7,
         func=lambda x: (x * k3x5).sum(),
          min_periods=7, center=True)
```